KAPASA MAKASA UNIVERSITY

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

CLOUD COMPUTING AND SERVICES (CYS 440)

Test 1 2026

Instructions

1. Answer all questions in hand written
2. Clearly indicate your full name, student id and the course code on all your answer scripts.
3. Scan (PDF ONLY) and send your answers to gathams@icloud.com
4. Submission after the stipulated time will not be accepted.

Section A (Answer All Questions)

1. Full virtualization means:

   a) The guest OS is modified to work with the hypervisor

   b) The complete hardware is simulated, allowing unmodified guest OSes to run

   c) Only part of the hardware is virtualized

   d) The hypervisor runs inside a virtual machine

2. The technique where all guest actions are performed through the monitor without the OS knowing is called:

   a) Partitioning

   b) Encapsulation

   c) Isolation

   d) Interposition

3. A developer wants to deploy a web application without managing the underlying operating system or runtime. Which service model is most appropriate?
   a) IaaS
   b) PaaS
   c) SaaS
   d) DaaS

4. A cloud that combines on-premises private cloud resources with public cloud services is called:
   a) Community cloud
   b) Multi-cloud
   c) Hybrid cloud
   d) Virtual private cloud

5. Which characteristic allows a user to provision server time and storage automatically without talking to a human at the provider?
   a) Broad network access
   b) On-demand self-service
   c) Resource pooling
   d) Measured service

6. The term “hypervisor” is also known as:
   a) Virtual machine monitor (VMM)
   b) Container engine
   c) Cloud orchestrator
   d) Load balancer

7. In a virtualized environment, what is the software layer that directly manages the hardware and schedules virtual machines called?
   a) Guest OS
   b) Virtual machine image
   c) Hypervisor (VMM)
   d) Cloud broker

8. Power Usage Effectiveness (PUE) is a metric used to measure:
   a) The processing speed of a data center
   b) The energy efficiency of a data center
   c) The number of virtual machines per server
   d) The network latency of a cloud service

9. Which service model provides virtual machines, raw storage, and virtual networks to customers?
   a) Software as a Service (SaaS)
   b) Platform as a Service (PaaS)
   c) Infrastructure as a Service (IaaS)
   d) Function as a Service (FaaS

10. According to the Uptime Institute Tier Classification, which Tier requires fault-tolerant (N+1) power and cooling paths and can sustain a 72-hour power outage?
    a) Tier I
    b) Tier II
    c) Tier III
    d) Tier IV

SECTION B (ANSWER ALL QUESTIONS)

1. Many people find it difficult to define cloud computing, in your own words define cloud computing.
2. Explain the concept of hot and cold aisles in a data center. Draw a simple diagram showing rack arrangement and airflow
3. Explain the five essential characteristics of cloud computing according to NIST.
4. Explain the difference between public cloud, private cloud, and hybrid cloud and community cloud.
5. List and explain at least four challenges of cloud computing that a cyber-security engineer must address.
6. Discuss any five main components of a data center.
7. Compare full virtualization and para virtualization. In which scenario would para virtualization be preferred?

8. Discuss with examples the cloud computing service models.
9. Distinguish bare metal from type two hypervisor.